

Absolute value

Outline

Definition

Triangle inequality

Definition

Introduction: Definition

- ▶ In this section we define the **absolute value** of a real number
- ▶ Geometrically, $|a|$ is the distance of a from 0 on the real line
- ▶ The absolute value is the one-dimensional prototype of the **norms** of vectors that we will study later
- ▶ We also see how to rewrite inequalities involving $|a|$ without the absolute value

Definition 1: absolute value

The **absolute value** $|a|$ of a real number a is the following non-negative real number:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases} \quad (1)$$

Inequalities with the absolute value

- ▶ From the definition of absolute value it immediately follows that:

$$|a| < \varepsilon \iff -\varepsilon < a < \varepsilon, \quad \forall \varepsilon > 0, a \in \mathbb{R} \quad (2)$$

- ▶ and, similarly, for non-strict inequalities:

$$|a| \leq \varepsilon \iff -\varepsilon \leq a \leq \varepsilon, \quad \forall \varepsilon \geq 0, a \in \mathbb{R} \quad (3)$$

Summary: Definition

Absolute value:

- ▶ $|a| = a$ if $a \geq 0$ and $|a| = -a$ if $a < 0$
- ▶ $|a|$ is always a non-negative real number

Inequalities:

- ▶ $|a| < \varepsilon \iff -\varepsilon < a < \varepsilon \quad (\varepsilon > 0)$
- ▶ $|a| \leq \varepsilon \iff -\varepsilon \leq a \leq \varepsilon \quad (\varepsilon \geq 0)$

Triangle inequality

Introduction: Triangle inequality

- ▶ The **triangle inequality** bounds the absolute value of a sum with the sum of the absolute values
- ▶ We prove it, derive some equivalent forms and the **reverse triangle inequality**
- ▶ It is one of the most used tools in analysis and optimization, and it will become one of the three axioms of a norm

Observation 1: triangle inequality

$$|b + c| \leq |b| + |c|, \quad \forall b, c \in \mathbb{R} \quad (4)$$

Proof

We write the two relations:

$$-|b| \leq b \leq |b|, \quad -|c| \leq c \leq |c|$$

and add member to member:

$$-(|b| + |c|) \leq b + c \leq |b| + |c|$$

So for (3), with $\varepsilon = |b| + |c| \geq 0$ and $a = b + c$, it follows (4).



Other forms of the triangle inequality (1/2)

- ▶ The triangle inequality is also used in the following form:

$$|d - e| \leq |d - f| + |e - f| \quad \forall d, e, f \in \mathbb{R} \quad (5)$$

- ▶ To obtain it, it suffices to set in (4):

$$b = d - f, \quad c = f - e$$

and we get:

$$|d - f + f - e| = |d - e| \leq |d - f| + |f - e| = |d - f| + |e - f|$$

since:

$$|f - e| = |e - f| \quad \forall e, f \in \mathbb{R}$$

Other forms of the triangle inequality (2/2)

- ▶ Finally, another form of the triangle inequality is:

$$|g| \leq |g-h| + |h| \quad \text{that is} \quad |g| - |h| \leq |g-h| \quad \forall g, h \in \mathbb{R} \quad (6)$$

- ▶ To obtain it, it suffices to set in (4):

$$b = g - h, \quad c = h$$

Observation 2: reverse triangle inequality

$$||g| - |h|| \leq |g - h|, \quad \forall g, h \in \mathbb{R} \quad (7)$$

Proof (1/2)

From (6), we have:

$$|g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R}$$

Similarly exchanging g for h in (6) we get:

$$|h| - |g| \leq |h - g| = |g - h| \quad \text{that is} \quad |g| - |h| \geq -|g - h|$$

Proof (2/2)

Then we have:

$$-(|g - h|) \leq |g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R}$$

Accordingly from (3), with $\varepsilon = |g - h| \geq 0$ and $a = |g| - |h|$, we get the reverse triangle inequality (7).



Further properties

- ▶ The triangle inequality (4) can easily be generalized to the case of k addends:

$$\left| \sum_{i=1}^k b_i \right| \leq \sum_{i=1}^k |b_i| \quad (8)$$

- ▶ The following immediate properties also hold, $\forall b, c \in \mathbb{R}$ ($c \neq 0$ in the quotient):

$$|b \, c| = |b| \, |c|, \quad \left| \frac{b}{c} \right| = \frac{|b|}{|c|}, \quad |-b| = |b| \quad (9)$$

Summary: Triangle inequality

Triangle inequality:

- ▶ $|b + c| \leq |b| + |c|$, and for k addends $\left| \sum_{i=1}^k b_i \right| \leq \sum_{i=1}^k |b_i|$
- ▶ Equivalent forms: $|d - e| \leq |d - f| + |e - f|$ and $||g| - |h|| \leq |g - h|$

Reverse triangle inequality:

- ▶ $||g| - |h|| \leq |g - h|$

Other properties:

- ▶ $|bc| = |b||c|$, $|b/c| = |b|/|c|$, $|-b| = |b|$

Key Takeaways

Summary

1. Definition:

- ▶ $|a|$ is a or $-a$, whichever is non-negative
- ▶ $|a| \leq \varepsilon$ is equivalent to the double inequality $-\varepsilon \leq a \leq \varepsilon$

2. Triangle inequality:

- ▶ $|b + c| \leq |b| + |c|$, proved by adding $-|b| \leq b \leq |b|$ and $-|c| \leq c \leq |c|$
- ▶ Reverse form: $||g| - |h|| \leq |g - h|$

3. Why it matters:

- ▶ These properties are the model for the axioms of norms on $\mathbb{R}^n$